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(54) **ANNULOPLASTY RING WITH POSTERIOR LEAFLET FOR MINIMALLY INVASIVE TREATMENT**

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,602,911 A 7/1986 Ahmadi et al.
4,874,378 A 10/1989 Hillstead

(Continued)

FOREIGN PATENT DOCUMENTS

CN 2114422 U 9/1992
CN 2633218 Y 8/2004

(Continued)

OTHER PUBLICATIONS

PCT/US2020/037294, International Search Report and Written Opinion, Aug. 28, 2020, 7 pages.

(Continued)

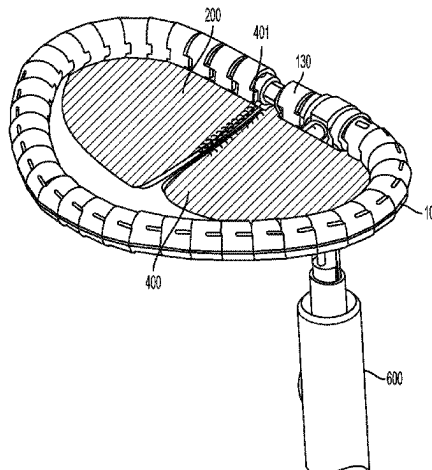
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(57) **ABSTRACT**

An annuloplasty ring is disclosed. The annuloplasty ring includes an outer tube, a plurality of anchors, and at least one synthetic leaflet. The synthetic leaflets are in mechanical communication with the annuloplasty ring at a plurality of points. The outer tube includes a plurality of windows. The plurality of anchors are positioned inside the outer tube and about a perimeter of the outer tube. The plurality of anchors are configured to be emitted from the plurality of windows in order to anchor the annuloplasty ring to a heart valve of a patient.

19 Claims, 17 Drawing Sheets



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(56)

References Cited

U.S. PATENT DOCUMENTS

4,953,540	A	9/1990	Ray et al.	11,806,237	B2	11/2023	Rozen et al.
5,080,662	A	1/1992	Paul	11,813,164	B2	11/2023	Yellin et al.
5,236,440	A	8/1993	Hlavacek	11,857,418	B2	1/2024	Yellin et al.
5,306,296	A	4/1994	Wright et al.	12,115,069	B2	10/2024	Shaolian et al.
5,609,565	A	3/1997	Nakamura	12,127,941	B2	10/2024	Yellin et al.
5,695,518	A	12/1997	Laerum	2002/0151961	A1	10/2002	Lashinski et al.
5,716,370	A	2/1998	Williamson, IV et al.	2002/0151970	A1	10/2002	Garrison et al.
5,855,614	A	1/1999	Stevens et al.	2002/0188170	A1	12/2002	Santamore et al.
6,113,611	A	9/2000	Allen et al.	2002/0198526	A1	12/2002	Shaolian et al.
6,231,602	B1	5/2001	Carpentier et al.	2003/0050649	A1	3/2003	Brock et al.
6,447,524	B1	9/2002	Knodel et al.	2003/0050693	A1	3/2003	Quijano et al.
6,619,291	B2	9/2003	Hlavka et al.	2003/0078465	A1	4/2003	Pai et al.
6,629,534	B1	10/2003	St. Goar et al.	2003/0078671	A1	4/2003	Lesniak et al.
6,669,687	B1	12/2003	Saadat	2003/0191528	A1	10/2003	Quijano et al.
6,689,048	B2	2/2004	Vanden Hoek et al.	2003/0198605	A1	10/2003	Montgomery
6,726,704	B1	4/2004	Loshakove et al.	2003/0199974	A1	10/2003	Lee et al.
6,776,784	B2	8/2004	Ginn	2004/0044364	A1	3/2004	DeVries et al.
6,790,229	B1	9/2004	Berrekouw	2004/0068276	A1	4/2004	Golden et al.
6,797,002	B2	9/2004	Spence et al.	2004/0073237	A1	4/2004	Leinsing
6,805,711	B2	10/2004	Quijano et al.	2004/0122514	A1	6/2004	Fogarty et al.
6,869,444	B2*	3/2005	Gabbay A61F 2/2466 128/898	2004/0138744	A1	7/2004	Lashinski et al.
6,893,459	B1	5/2005	Macoviak	2004/0148021	A1	7/2004	Cartledge et al.
7,101,395	B2	9/2006	Tremulis et al.	2004/0193191	A1	9/2004	Starksen et al.
7,114,953	B1	10/2006	Wagner	2004/0243230	A1	12/2004	Navia et al.
7,175,660	B2	2/2007	Cartledge et al.	2004/0249391	A1	12/2004	Cummins
7,238,191	B2	7/2007	Bachmann	2004/0260393	A1	12/2004	Rahdert et al.
7,285,087	B2	10/2007	Moaddeb et al.	2004/0260394	A1	12/2004	Douk et al.
7,297,150	B2	11/2007	Cartledge et al.	2005/0004668	A1	1/2005	Aklog et al.
7,569,072	B2	8/2009	Berg et al.	2005/0020696	A1	1/2005	Montgomery et al.
7,594,887	B2	9/2009	Moaddeb et al.	2005/0033325	A1	2/2005	May et al.
7,635,329	B2	12/2009	Goldfarb et al.	2005/0065550	A1	3/2005	Starksen et al.
7,655,040	B2	2/2010	Douk et al.	2005/0090846	A1	4/2005	Pedersen et al.
7,717,954	B2	5/2010	Solem et al.	2005/0096740	A1	5/2005	Langberg et al.
7,722,668	B2	5/2010	Moaddeb et al.	2005/0113910	A1	5/2005	Paniagua et al.
7,758,637	B2	7/2010	Starksen et al.	2005/0137692	A1	6/2005	Haug et al.
7,828,819	B2	11/2010	Webler et al.	2005/0137695	A1	6/2005	Salahieh et al.
7,837,729	B2	11/2010	Gordon et al.	2005/0203549	A1	9/2005	Realyvasquez
7,988,725	B2	8/2011	Gross et al.	2005/0222678	A1	10/2005	Lashinski et al.
8,163,014	B2	4/2012	Lane et al.	2005/0240200	A1	10/2005	Bergheim
8,182,529	B2	5/2012	Gordon et al.	2005/0250161	A1	11/2005	Suciu-Foca et al.
8,236,049	B2	8/2012	Rowe et al.	2005/0267572	A1	12/2005	Schoon et al.
8,287,591	B2	10/2012	Keidar et al.	2005/0283190	A1	12/2005	Huitema et al.
8,518,107	B2	8/2013	Tsukashima et al.	2005/0288778	A1	12/2005	Shaoulian et al.
8,579,968	B1	11/2013	Shannon et al.	2005/0288781	A1	12/2005	Moaddeb et al.
8,690,939	B2	4/2014	Miller et al.	2006/0009737	A1	1/2006	Whiting et al.
8,821,570	B2	9/2014	DuMontelle et al.	2006/0020327	A1	1/2006	Lashinski et al.
9,180,008	B2	11/2015	Yellin et al.	2006/0020334	A1*	1/2006	Lashinski A61F 2/2418 623/2.11
9,402,721	B2	8/2016	Buchbinder et al.	2006/0052867	A1	3/2006	Revuelta et al.
9,433,503	B2	9/2016	Tsukashima et al.	2006/0122633	A1	6/2006	To et al.
9,839,519	B2	12/2017	Shaolian et al.	2006/0129025	A1	6/2006	Levine et al.
9,877,833	B1	1/2018	Bishop et al.	2006/0155165	A1	7/2006	Vanden Hoek et al.
10,405,979	B2	9/2019	Schaffner et al.	2006/0161169	A1	7/2006	Nieminen et al.
10,543,087	B2	1/2020	Yellin et al.	2006/0184240	A1	8/2006	Jimenez et al.
10,779,945	B2	9/2020	Buchbinder et al.	2006/0184242	A1	8/2006	Lichtenstein
11,058,417	B2	7/2021	Foerster et al.	2006/0195134	A1	8/2006	Crittenden
11,191,536	B2	12/2021	Foerster et al.	2006/0195183	A1	8/2006	Navia et al.
11,224,422	B2	1/2022	Foerster et al.	2006/0241748	A1	10/2006	Lee et al.
11,298,230	B2	4/2022	Shaolian et al.	2006/0282161	A1	12/2006	Huynh et al.
11,382,749	B2	7/2022	Yellin et al.	2007/0016287	A1	1/2007	Cartledge et al.
11,510,835	B2	11/2022	Yellin et al.	2007/0027533	A1	2/2007	Douk
11,534,300	B2	12/2022	Yellin et al.	2007/0038296	A1	2/2007	Navia
11,571,301	B2	2/2023	Yellin et al.	2007/0051377	A1	3/2007	Douk et al.
11,571,307	B2	2/2023	Yellin et al.	2007/0067027	A1	3/2007	Moaddeb et al.
11,576,779	B2	2/2023	Yellin et al.	2007/0073098	A1	3/2007	Lenker et al.
11,617,647	B2	4/2023	Yellin	2007/0080188	A1	4/2007	Spence et al.
11,654,018	B2	5/2023	Shaolian et al.	2007/0093854	A1	4/2007	Kayan
11,793,628	B2	10/2023	Dumontelle et al.	2007/0118215	A1	5/2007	Moaddeb
11,806,009	B2	11/2023	Foerster et al.	2007/0128132	A1	6/2007	Piergallini et al.
				2007/0135913	A1	6/2007	Moaddeb et al.
				2007/0142907	A1	6/2007	Moaddeb et al.
				2007/0213812	A1	9/2007	Webler et al.
				2007/0233239	A1	10/2007	Navia et al.
				2007/0239272	A1	10/2007	Navia et al.
				2007/0244553	A1	10/2007	Rafiee et al.
				2007/0244554	A1	10/2007	Rafiee et al.
				2007/0244555	A1	10/2007	Rafiee et al.
				2007/0244556	A1	10/2007	Rafiee et al.
				2007/0250161	A1	10/2007	Dolan

(56)

References Cited

U.S. PATENT DOCUMENTS

2007/0293942	A1	12/2007	Mirzaee	2013/0087598	A1	4/2013	Surti
2008/0039934	A1	2/2008	Styrz	2013/0116780	A1	5/2013	Miller et al.
2008/0051807	A1	2/2008	St Goar et al.	2013/0166022	A1	6/2013	Conklin
2008/0065011	A1	3/2008	Marchand et al.	2013/0204361	A1	8/2013	Adams et al.
2008/0177380	A1	7/2008	Starksen et al.	2013/0226289	A1	8/2013	Shaolian et al.
2008/0177381	A1	7/2008	Navia et al.	2013/0226290	A1	8/2013	Yellin et al.
2008/0200980	A1	8/2008	Robin et al.	2013/0282114	A1	10/2013	Schweich, Jr. et al.
2008/0208327	A1	8/2008	Rowe	2013/0289718	A1	10/2013	Tsukashima et al.
2008/0215145	A1	9/2008	Moaddeb et al.	2013/0289720	A1	10/2013	Dobrilovic
2008/0243220	A1	10/2008	Barker	2013/0304197	A1	11/2013	Buchbinder et al.
2008/0262513	A1	10/2008	Stahler et al.	2014/0005778	A1	1/2014	Buchbinder et al.
2008/0262609	A1	10/2008	Gross et al.	2014/0025163	A1	1/2014	Padalla et al.
2008/0288060	A1	11/2008	Kaye et al.	2014/0046433	A1	2/2014	Kovalsky
2008/0306586	A1	12/2008	Cartledge et al.	2014/0058505	A1	2/2014	Bielefeld
2009/0076599	A1	3/2009	Bergin	2014/0114407	A1	4/2014	Rajamannan
2009/0088838	A1	4/2009	Shaolian et al.	2014/0188130	A1	7/2014	Sanchez et al.
2009/0118747	A1	5/2009	Bettuchi et al.	2014/0214157	A1	7/2014	Börtlein et al.
2009/0125098	A1	5/2009	Chuter	2014/0277411	A1	9/2014	Börtlein et al.
2009/0149872	A1	6/2009	Gross et al.	2014/0364939	A1	12/2014	Deshmukh et al.
2009/0216322	A1	8/2009	Le et al.	2015/0073420	A1	3/2015	Bookwalter et al.
2009/0222083	A1	9/2009	Nguyen et al.	2015/0112432	A1	4/2015	Reich et al.
2009/0238778	A1	9/2009	Mordas et al.	2015/0127093	A1	5/2015	Hosmer et al.
2009/0299470	A1	12/2009	Rao et al.	2015/0173897	A1	6/2015	Raanani et al.
2010/0010616	A1	1/2010	Drews et al.	2015/0173987	A1	6/2015	Albinmoussa et al.
2010/0030014	A1	2/2010	Ferrazzi	2015/0351903	A1	12/2015	Morris et al.
2010/0063586	A1	3/2010	Hasenkam et al.	2015/0366666	A1	12/2015	Khairkahan et al.
2010/0076549	A1	3/2010	Keidar et al.	2016/0022419	A1	1/2016	Yellin et al.
2010/0121433	A1	5/2010	Bolling et al.	2016/0038286	A1	2/2016	Yellin et al.
2010/0161047	A1	6/2010	Cabiri	2016/0089235	A1	3/2016	Yellin
2010/0185274	A1	7/2010	Moaddeb et al.	2016/0100897	A1	4/2016	Avalos et al.
2010/0191327	A1	7/2010	Lane et al.	2016/0106420	A1	4/2016	Foerster et al.
2010/0204713	A1	8/2010	Ruiz Morales	2016/0120642	A1	5/2016	Shaolian et al.
2010/0211166	A1	8/2010	Miller et al.	2016/0120645	A1	5/2016	Alon
2010/0249920	A1	9/2010	Bolling et al.	2016/0220371	A1	8/2016	Keane
2010/0262232	A1	10/2010	Annest	2016/0270911	A1	9/2016	Ganesan et al.
2010/0266989	A1	10/2010	Piergallini et al.	2016/0331534	A1	11/2016	Buchbinder et al.
2010/0280604	A1	11/2010	Zipory et al.	2017/0042670	A1	2/2017	Shaolian et al.
2010/0280605	A1	11/2010	Hammer et al.	2017/0231763	A1	8/2017	Yellin
2010/0286767	A1	11/2010	Zipory et al.	2017/0258590	A1	9/2017	Khairkahan
2011/0004299	A1	1/2011	Navia et al.	2017/0296340	A1	10/2017	Gross et al.
2011/0011917	A1	1/2011	Loulmet	2017/0325948	A1	11/2017	Wallace et al.
2011/0022168	A1	1/2011	Cartledge	2018/0028387	A1	2/2018	Yellin et al.
2011/0027753	A1	2/2011	Maurat et al.	2018/0042723	A1	2/2018	Yellin et al.
2011/0034953	A1	2/2011	Milo	2018/0098849	A1	4/2018	Yellin et al.
2011/0034999	A1	2/2011	Carpentier et al.	2018/0161160	A1	6/2018	Shaolian et al.
2011/0060407	A1	3/2011	Ketai et al.	2018/0161161	A1	6/2018	Yellin et al.
2011/0066231	A1	3/2011	Cartledge et al.	2018/0228610	A1	8/2018	Lashinski et al.
2011/0093062	A1	4/2011	Cartledge et al.	2018/0235758	A1*	8/2018	Biadillah A61F 2/246
2011/0106245	A1	5/2011	Miller et al.	2018/0325670	A1	11/2018	De
2011/0106247	A1	5/2011	Miller et al.	2019/0053905	A1	2/2019	Alon
2011/0137397	A1	6/2011	Chau et al.	2019/0083091	A1	3/2019	Foerster et al.
2011/0166649	A1	7/2011	Gross et al.	2019/0083092	A1	3/2019	Foerster et al.
2011/0190879	A1	8/2011	Bobo et al.	2019/0083239	A1	3/2019	Shaolian et al.
2011/0208295	A1	8/2011	Cartledge et al.	2019/0083240	A1*	3/2019	Shaolian A61F 2/2409
2011/0208298	A1	8/2011	Tuval et al.	2019/0091022	A1	3/2019	Yellin et al.
2011/0224785	A1	9/2011	Hacohen	2019/0133765	A1	5/2019	Yellin et al.
2011/0257728	A1	10/2011	Kuehn	2020/0069426	A1	3/2020	Conklin et al.
2011/0282361	A1	11/2011	Miller et al.	2020/0163763	A1	5/2020	Zipory et al.
2011/0301698	A1	12/2011	Miller et al.	2020/0170799	A1	6/2020	Yellin et al.
2011/0301699	A1	12/2011	Saadat	2020/0330228	A1	10/2020	Anderson et al.
2011/0319989	A1	12/2011	Lane et al.	2021/0085463	A1	3/2021	Yellin et al.
2012/0022557	A1	1/2012	Cabiri et al.	2021/0161662	A1	6/2021	Albes
2012/0022639	A1	1/2012	Hacohen et al.	2021/0346159	A1*	11/2021	Keränen A61F 2/2448
2012/0022644	A1	1/2012	Reich et al.	2021/0353417	A1	11/2021	Yellin et al.
2012/0059458	A1	3/2012	Buchbinder et al.	2022/0226771	A1	7/2022	Lipscomb
2012/0083880	A1	4/2012	Rankin et al.	2023/0040083	A1	2/2023	Gifford, III et al.
2012/0095455	A1	4/2012	Rodmond et al.	2023/0045532	A1	2/2023	Galler et al.
2012/0123531	A1*	5/2012	Tsukashima A61F 2/2448	2023/0372086	A1	11/2023	Galler et al.
			623/2.37				
2012/0136436	A1	5/2012	Cabiri et al.				
2012/0136463	A1	5/2012	Muniz				
2012/0165930	A1	6/2012	Gifford, III et al.				
2012/0245604	A1	9/2012	Tegzes				
2012/0310330	A1	12/2012	Buchbinder et al.				
2013/0066342	A1	3/2013	Dell et al.				

FOREIGN PATENT DOCUMENTS

CN	101411632	A	4/2009
CN	101460113	A	6/2009
CN	101553190	A	10/2009
CN	102014797	A	4/2011
CN	102088930	A	6/2011
CN	202859386	U	4/2013
CN	103179920	A	6/2013
CN	103237523	A	8/2013

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN	103735337	A	4/2014
CN	203954080	U	11/2014
CN	108618871	A	10/2018
CN	113855324	A	12/2021
DE	102014102653	A1	9/2015
EP	1752115	A1	2/2007
EP	2471464	A1	7/2012
EP	2600799	A2	6/2013
EP	2928538	A1	10/2015
EP	2967700	A1	1/2016
EP	2600799	B1	5/2017
EP	3213715	A1	9/2017
EP	2928538	B1	11/2018
FR	2845889	A1	4/2004
GB	1496804	A	1/1978
GB	2366319	A	3/2002
KR	20040095482	A	11/2004
RU	125062	U1	2/2013
WO	WO-8000673	A1	4/1980
WO	WO-9009153	A1	8/1990
WO	WO-9728745	A1	8/1997
WO	WO-03017874	A1	3/2003
WO	WO-03047467	A1	6/2003
WO	WO-2005046488	A2	5/2005
WO	WO-2007035882	A2	3/2007
WO	WO-2008097999	A2	8/2008
WO	WO-2009052427	A1	4/2009
WO	WO-2009120764	A2	10/2009
WO	WO-2010004546	A1	1/2010
WO	WO-2010085659	A1	7/2010
WO	WO-2011011443	A2	1/2011
WO	WO-2011097355	A2	8/2011
WO	WO-2011154942	A2	12/2011
WO	WO-2012004679	A2	1/2012
WO	WO-2012019052	A2	2/2012
WO	WO-2012038550	A1	3/2012
WO	WO-2012040865	A1	4/2012
WO	WO-2012063228	A1	5/2012
WO	WO-2012095159	A2	7/2012
WO	WO-2012106354	A1	8/2012
WO	WO-2012167095	A2	12/2012
WO	WO-2013095816	A1	6/2013
WO	WO-2013128436	A1	9/2013
WO	WO-2013130641	A1	9/2013
WO	WO-2013175468	A2	11/2013
WO	WO-2014089424	A1	6/2014
WO	WO-2014145399	A1	9/2014
WO	WO-2014178869	A1	11/2014
WO	WO-2014189509	A1	11/2014
WO	WO-2014190329	A1	11/2014
WO	WO-2014210600	A2	12/2014
WO	WO-2015052629	A1	4/2015
WO	WO-2015132668	A1	9/2015
WO	WO-2016025894	A1	2/2016
WO	WO-2016040526	A1	3/2016
WO	WO-2018035118	A1	2/2018
WO	WO-2018071540	A1	4/2018
WO	WO-2018170424	A1	9/2018
WO	WO-2020117842	A1	6/2020
WO	WO-2020252200	A1	12/2020

OTHER PUBLICATIONS

Extended European Search Report for European Application No. 11815347.7, mailed Mar. 14, 2016, 10 Pages.

Extended European Search Report for European Application No. 12793292.9, mailed Dec. 1, 2014, 6 Pages.

Extended European Search Report for European Application No. 13755441.6, mailed Mar. 1, 2016, 12 Pages.

Extended European Search Report for European Application No. 13860442.6, mailed Aug. 11, 2016, 7 pages.

Extended European Search Report for European Application No. 13885021.9, mailed Jan. 3, 2017, 8 Pages.

Extended European Search Report for European Application No. 14762806.9, mailed Jul. 29, 2016, 7 Pages.

Extended European Search Report for European Application No. 14801009.3, mailed Dec. 5, 2016, 8 Pages.

Extended European Search Report for European Application No. 14817662.1, mailed Jan. 23, 2017, 7 Pages.

Extended European Search Report for European Application No. 17155803.4, mailed Aug. 9, 2017, 10 Pages.

Extended European Search Report for European Application No. 17835256.3, mailed Feb. 12, 2020, 9 Pages.

Extended European Search Report for European Application No. 17841988.3, mailed Dec. 16, 2019, 8 Pages.

Extended European Search Report for European Application No. 17860901.2, mailed Jun. 5, 2020, 06 Pages.

Extended European Search Report for European Application No. 18768197.8, mailed Oct. 19, 2020, 7 Pages.

Extended European Search Report for European Application No. 19151726.7, mailed Jul. 22, 2019, 9 Pages.

Extended European Search Report for European Application No. 19170261.2, mailed Aug. 5, 2019, 9 pages.

Extended European Search Report for European Application No. 19893113.1, mailed Nov. 17, 2022, 7 Pages.

Extended European Search Report for European Application No. 20206790.6, mailed Dec. 7, 2020, 9 Pages.

Extended European Search Report for European Application No. 20209605.3, mailed Mar. 9, 2021, 7 pages.

Extended European Search Report for European Application No. 20823198.5, mailed May 15, 2023, 15 Pages.

Extended European Search Report for European Application No. 20823682.8, mailed Apr. 14, 2023, 10 Pages.

Extended European Search Report for European Application No. 20841346.8, mailed Jul. 21, 2023, 9 Pages.

International Search Report & Written Opinion dated Jul. 24, 2024 in Int'l PCT Patent Appl. Serial No. PCT/US2024/019797.

International Search Report and Written Opinion for International Application No. PCT/IL2022/050868, mailed Nov. 17, 2022, 19 Pages.

International Search Report and Written Opinion for International Application No. PCT/IL2023/050527, mailed Aug. 8, 2023, 14 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2011/046659, mailed Jun. 4, 2012, 13 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2012/040481, mailed Dec. 6, 2012, 12 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2013/028065, mailed Jun. 27, 2013, 11 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2013/042275, mailed Feb. 20, 2014, 18 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2013/058102, mailed Apr. 21, 2014, 10 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2013/073552, mailed Mar. 6, 2014, 10 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2014/030163, mailed Aug. 27, 2014, 10 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2014/039454, mailed Oct. 22, 2014, 11 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2014/044920, mailed Dec. 24, 2014, 15 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2017/044129, mailed Sep. 27, 2017, 9 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2017/046933, mailed Dec. 21, 2017, 11 Pages.

(56)

References Cited

OTHER PUBLICATIONS

International Search Report and Written Opinion for International Application No. PCT/US2017/056138, mailed Jan. 8, 2018 , 6 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2018/022910, mailed May 23, 2018, 7 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2019/064289, mailed Feb. 5, 2020, 8 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2020/037296, mailed Sep. 10, 2020, 8 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2020/042201, mailed Oct. 9, 2020, 7 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2021/071467, mailed Jan. 14, 2022, 12 Pages.

International Search Report and Written Opinion for International Application No. PCT/US2021/071468, mailed Jan. 19, 2022, 8 Pages.

Partial Supplementary European Search Report for European Application No. 11815347.7, mailed Nov. 16, 2015, 06 Pages.

Partial Supplementary European Search Report for European Application No. 13755441.6, mailed Nov. 3, 2015, 7 Pages.

* cited by examiner

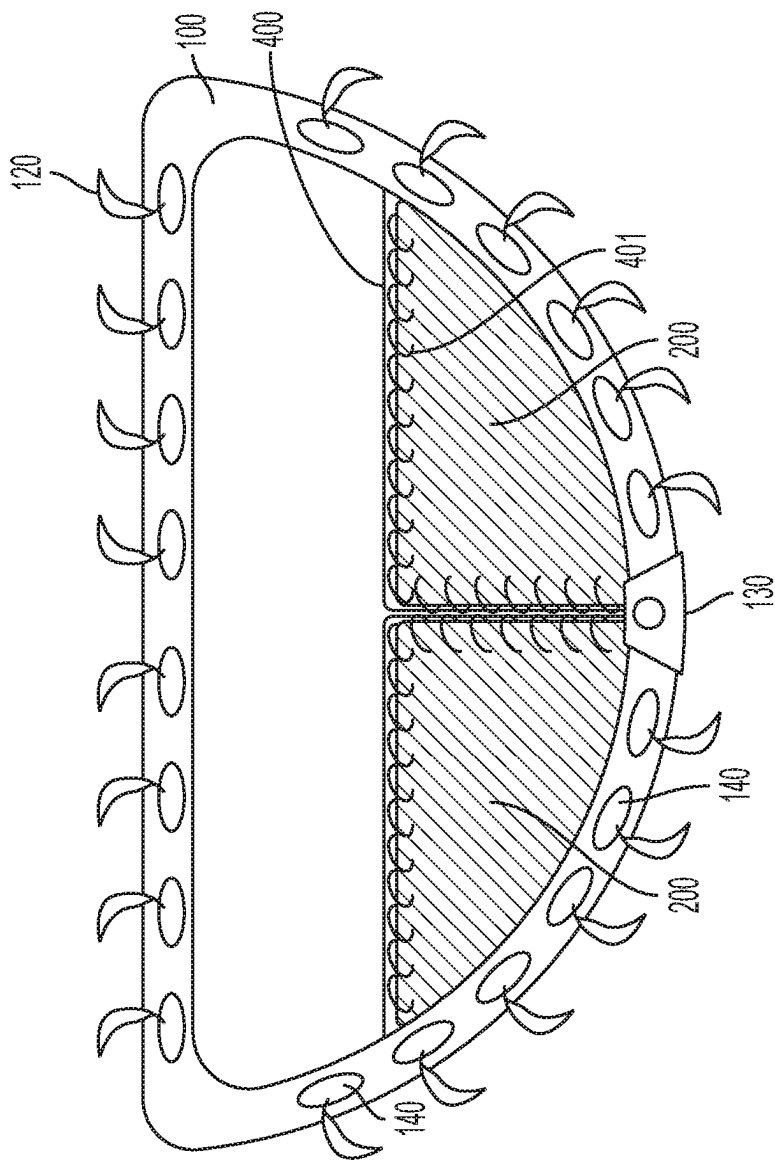


FIG. 1

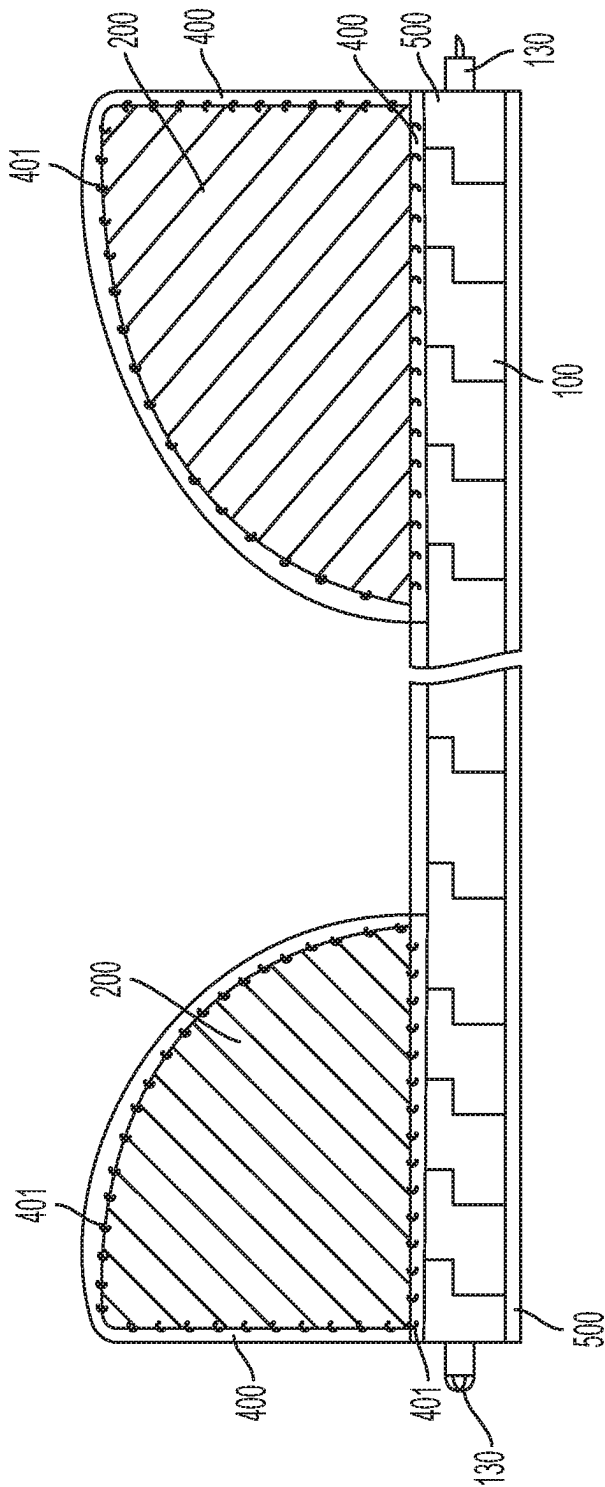


FIG. 2

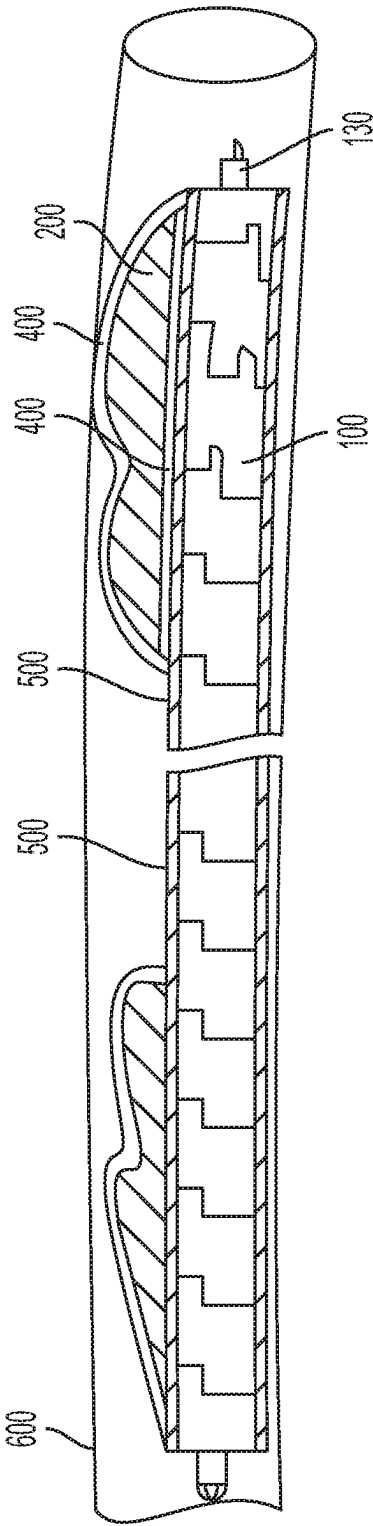
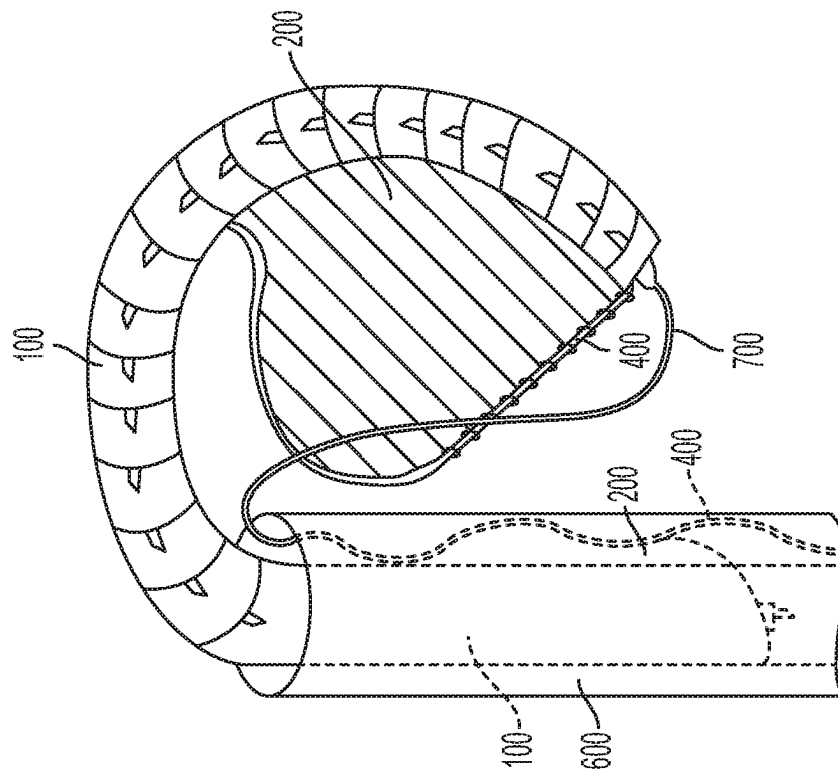


FIG. 3



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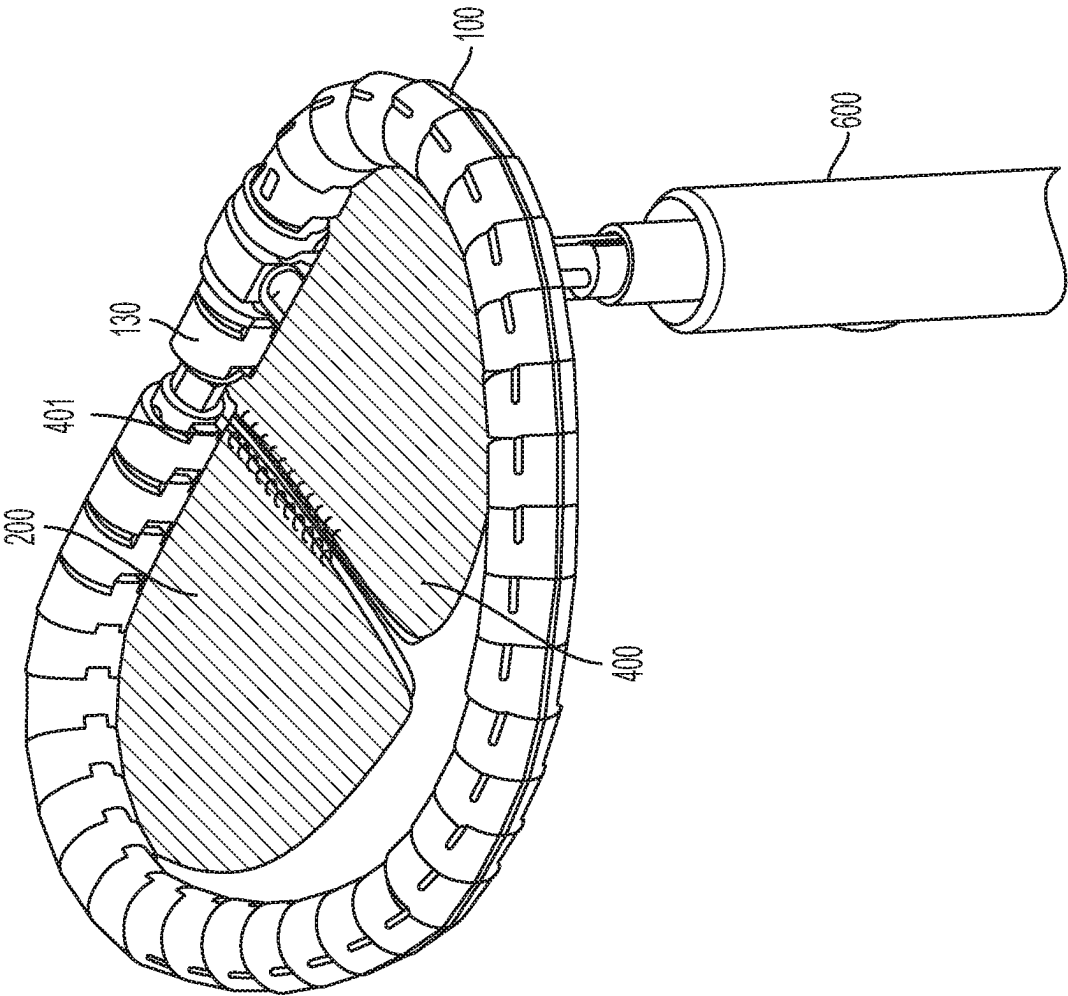


FIG. 5A

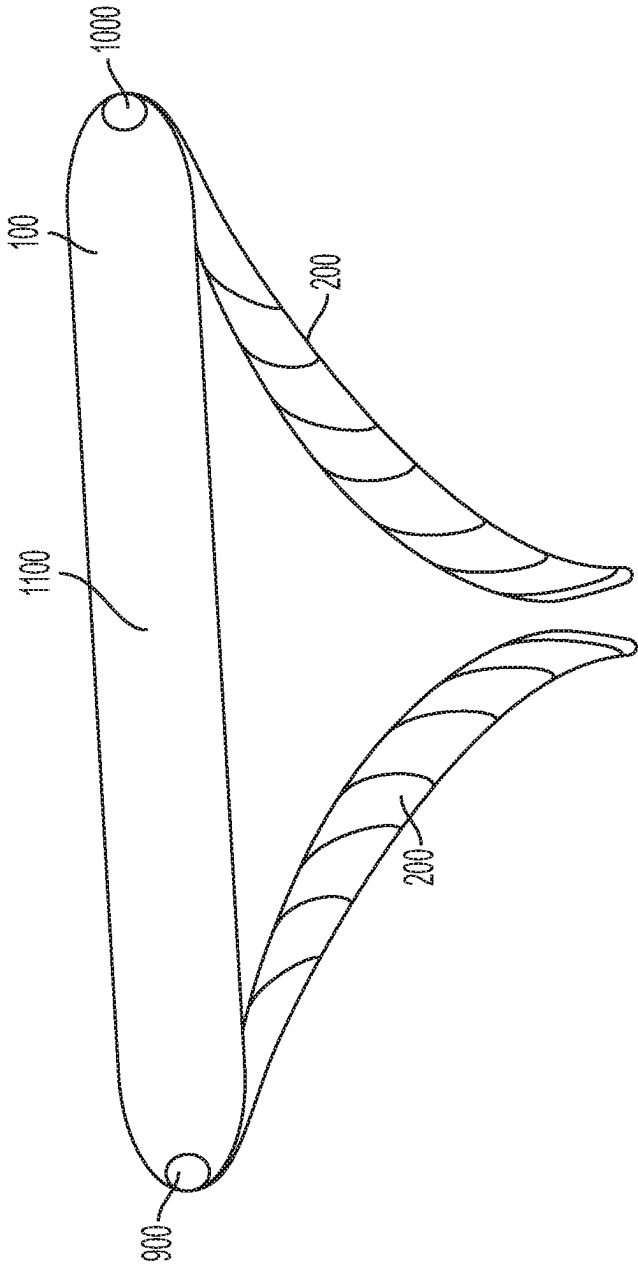


FIG. 5B

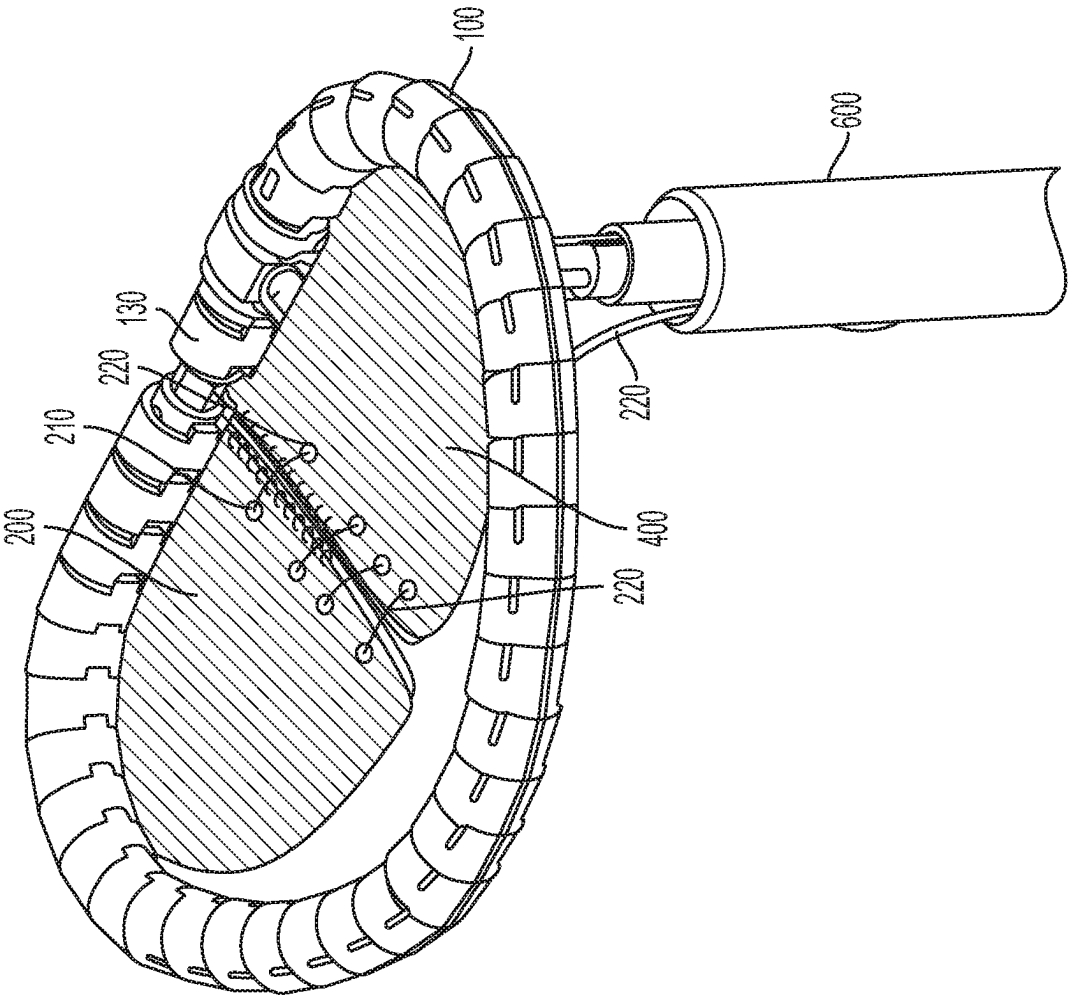


FIG. 5C

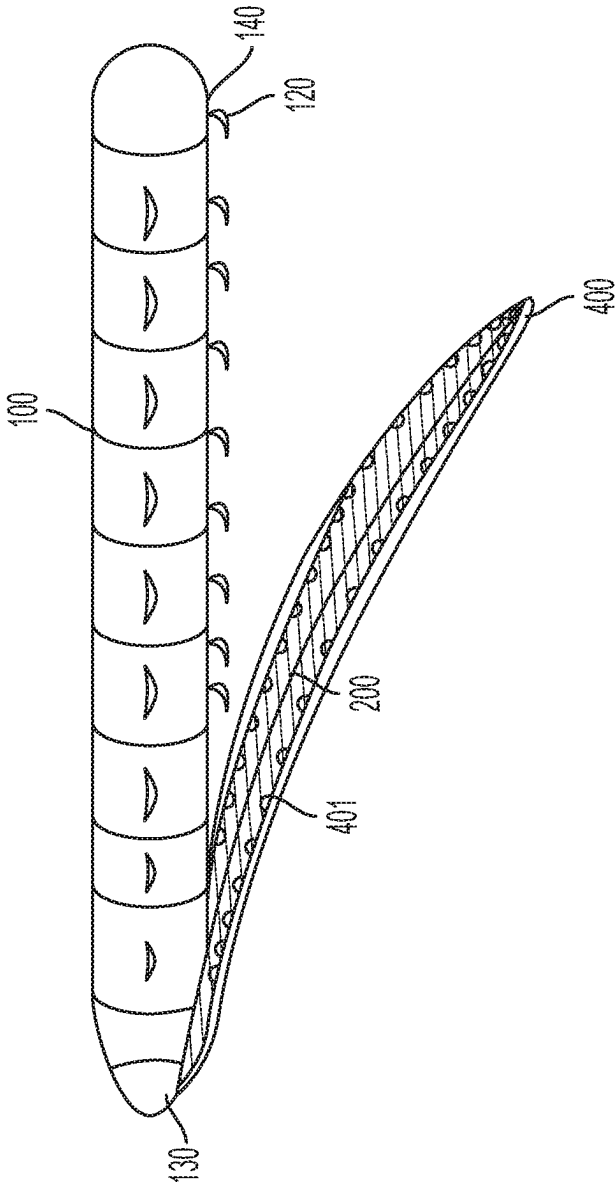


FIG. 6

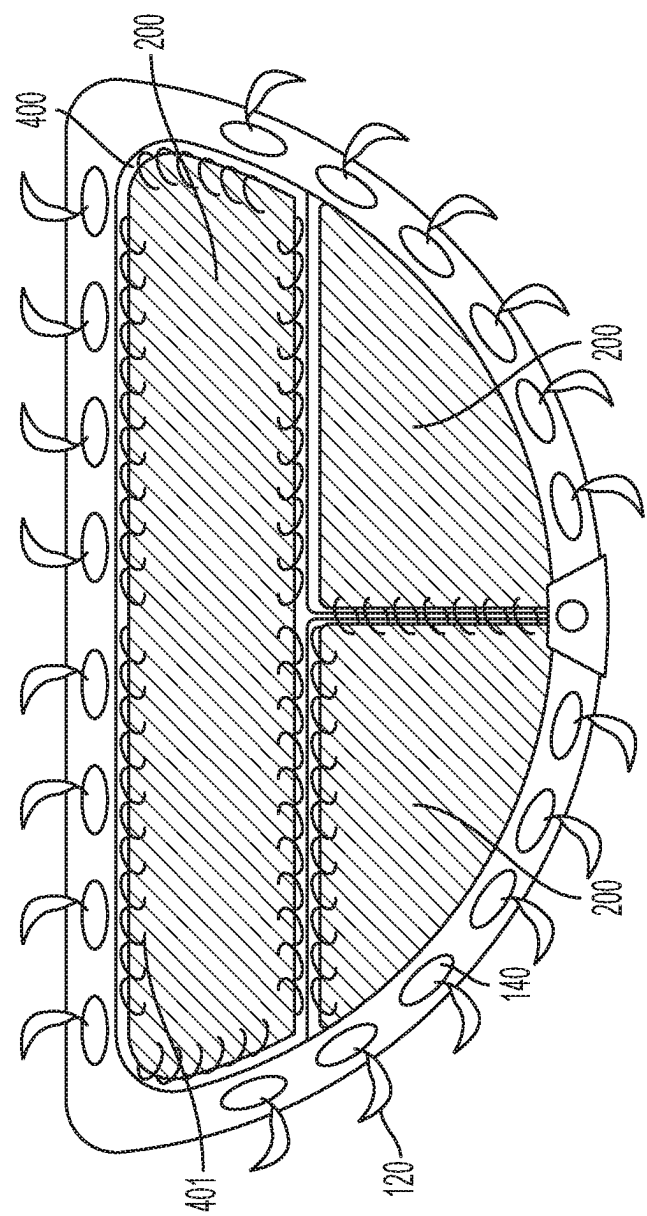
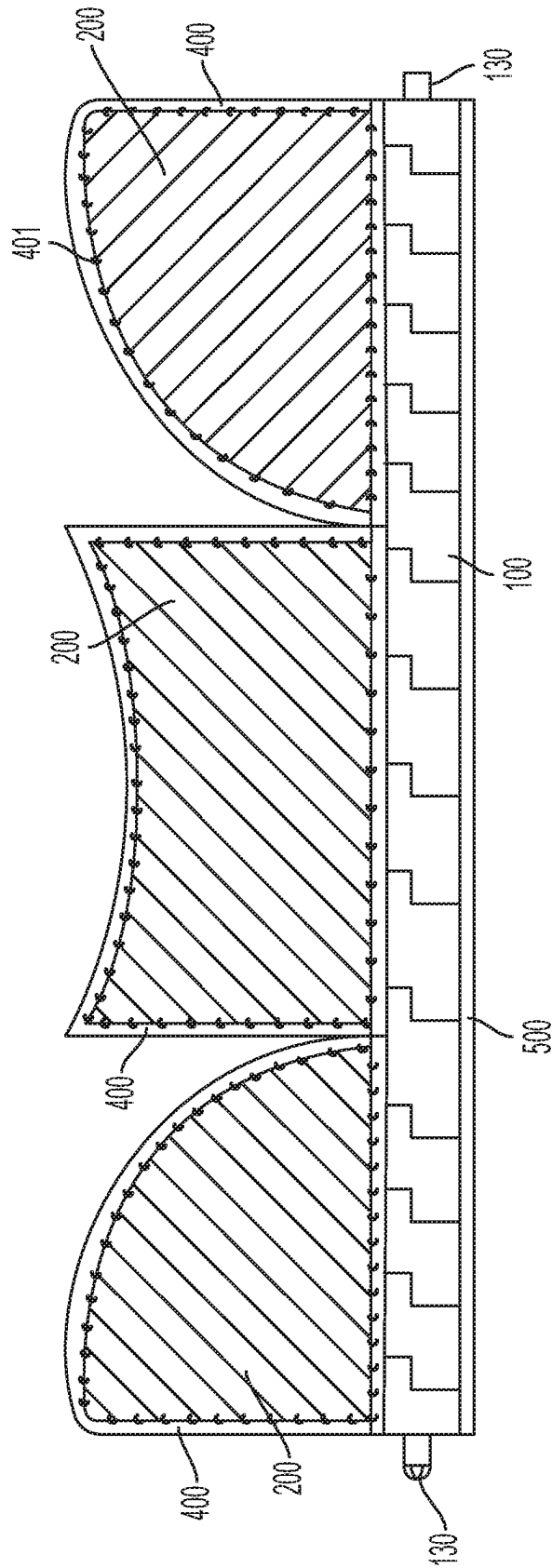


FIG. 7



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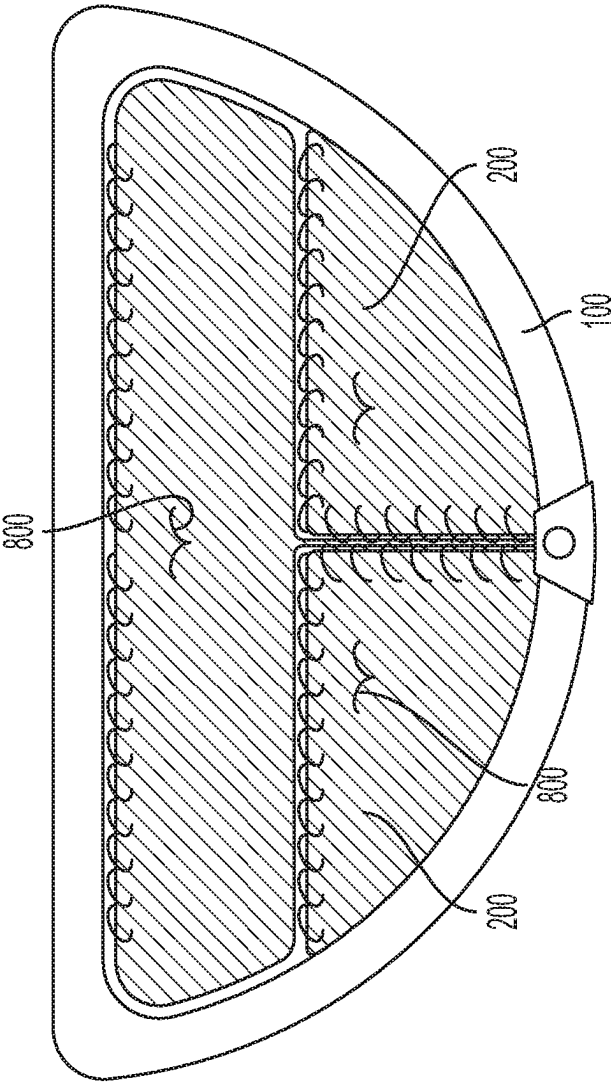


FIG. 9

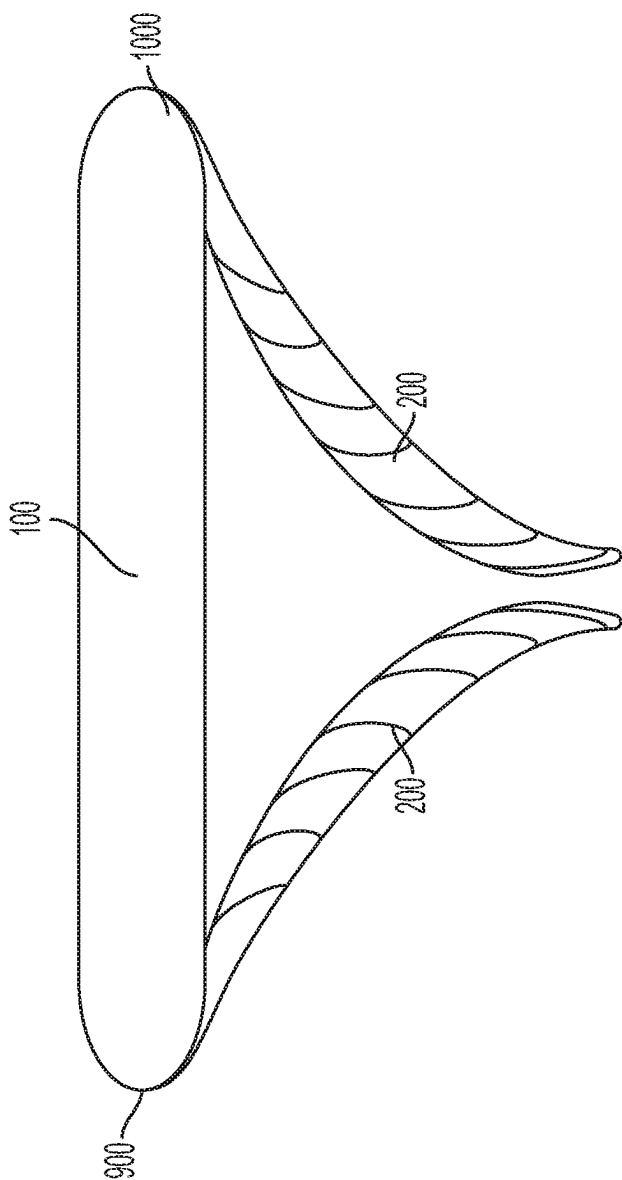


FIG. 10

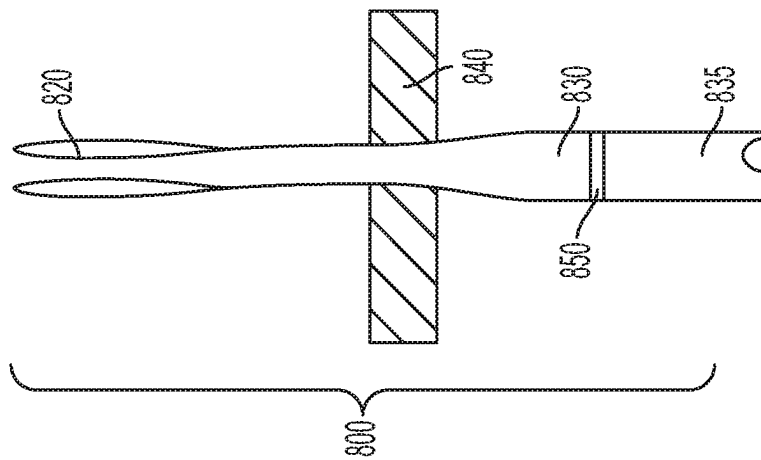


FIG. 11A

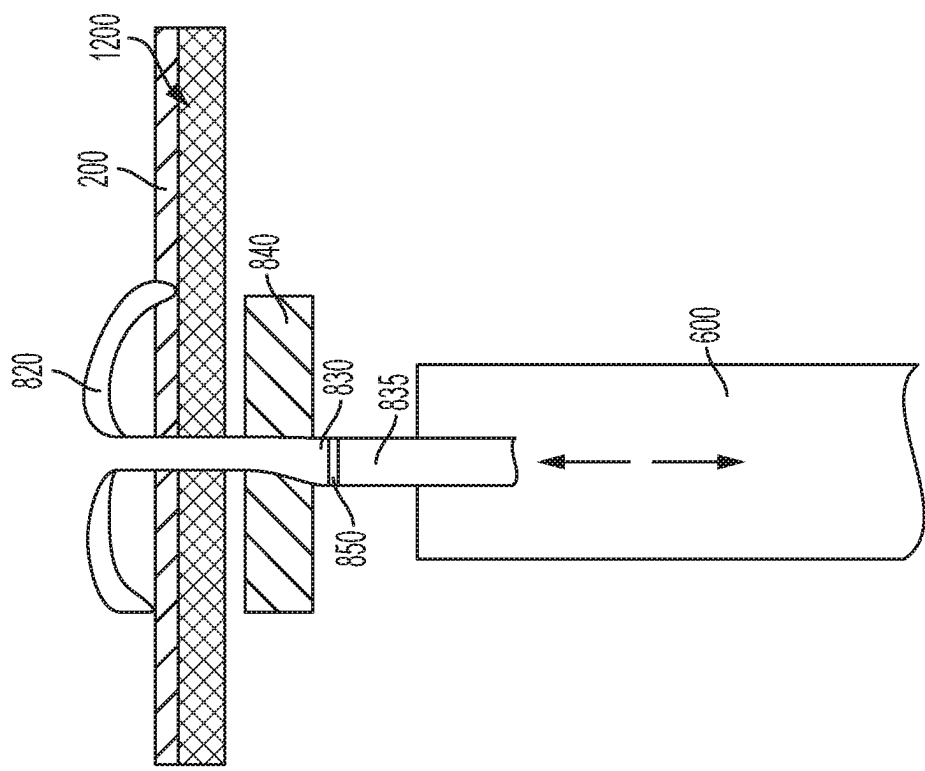


FIG. 11B

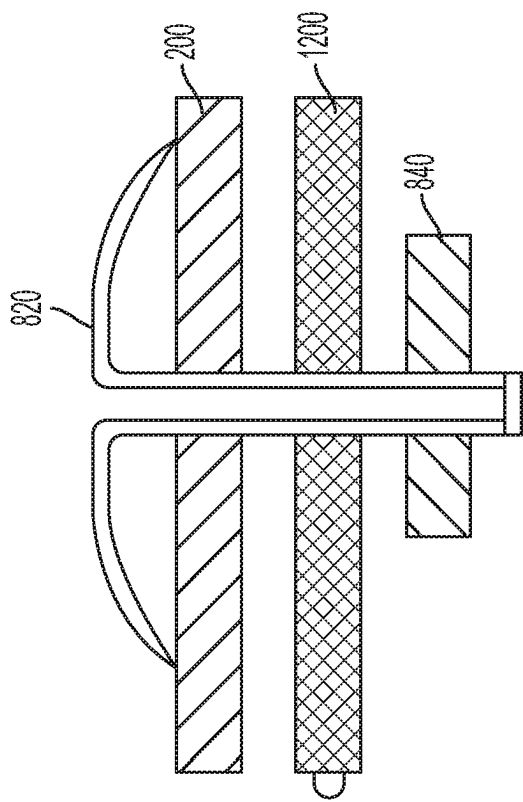


FIG. 11C

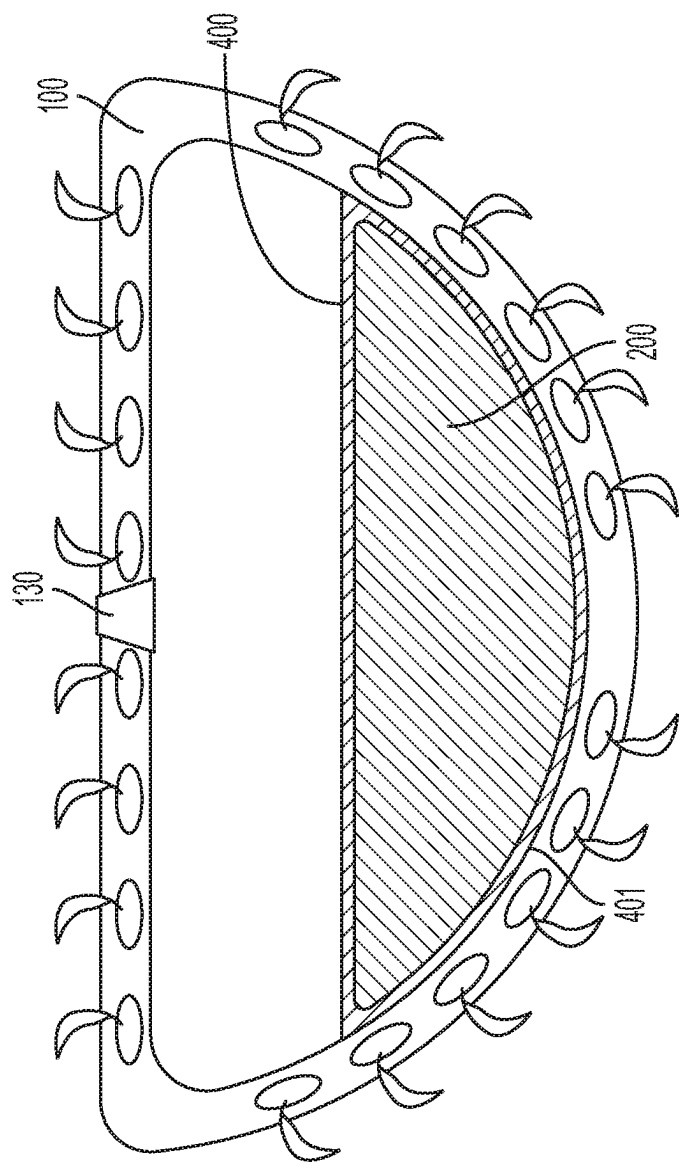


FIG. 12

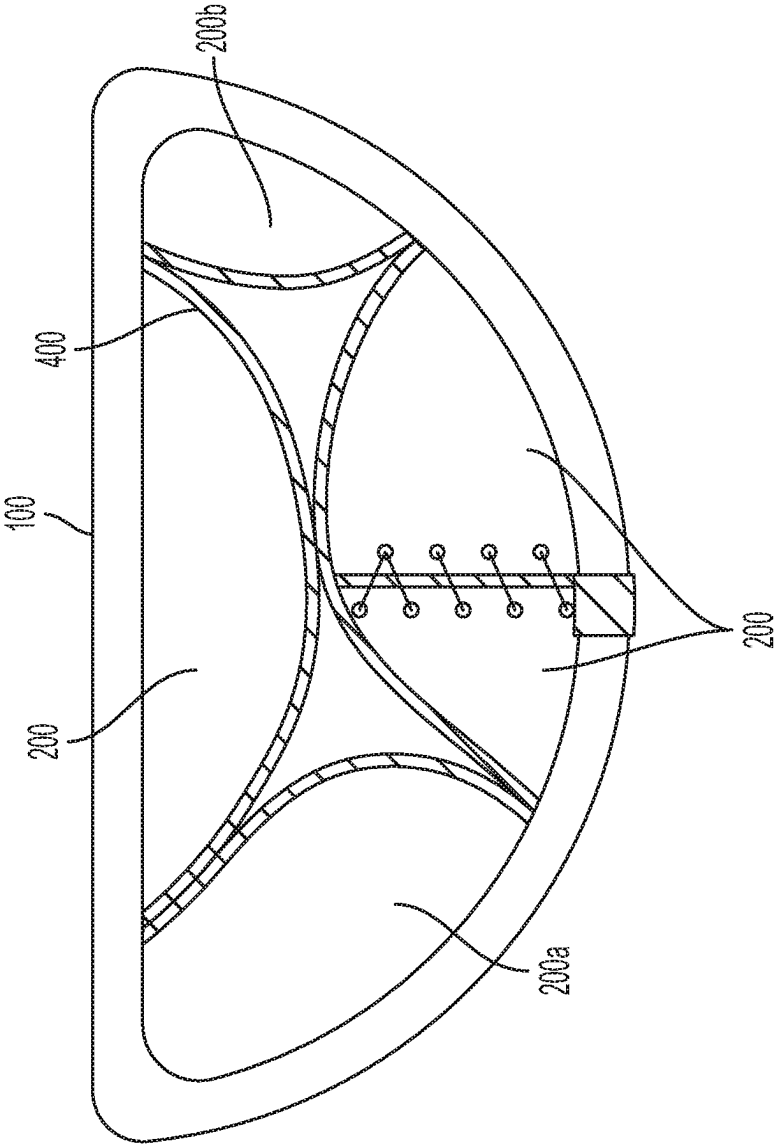


FIG. 13

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ANNULOPLASTY RING WITH POSTERIOR LEAFLET FOR MINIMALLY INVASIVE TREATMENT

CLAIM OF PRIORITY

The present application is a national stage filing under 35 U.S.C. § 371 of International Patent Application No. PCT/US2020/037294, filed Jun. 11, 2020, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/859,975, filed on Jun. 11, 2019, titled “ANNULOPLASTY RING WITH POSTERIOR LEAFLET FOR MINIMALLY INVASIVE TREATMENT,” each of which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

The present disclosure relates to implantable prosthetic devices. More specifically, the disclosure is directed to an improved prosthetic device implantable by catheter for the treatment of mitral or tricuspid regurgitation.

BACKGROUND

Mitral regurgitation is a valvular dysfunction that causes blood volume to flow during systole (during left ventricular contraction) from the left ventricle to the left atrium as opposed to a healthy heart where this direction of flow is blocked by the mitral valve. The reverse flow during systole causes pressure to rise in the left atrium. Maintaining a normal cardiac output results in an increased left ventricle pressure.

Treating patients with mitral regurgitation (MR) or tricuspid regurgitation (TR) could require valve replacement in order to reduce or eliminate the regurgitation. For many years, the commonly accepted treatment was surgical repair or replacement of the native valve during open heart surgery. In recent years, a trans-vascular technique has been developed for introducing and implanting a prosthetic heart valve using a flexible catheter in a manner that is less invasive than open heart surgery.

In the trans-vascular technique, the prosthetic is delivered to the target site (aortic valve, mitral valve, tricuspid valve, or other valve) through a catheter while the device is crimped to a low diameter shaft, and expanded/deployed to the functional size when it is located in the correct position.

The catheter can be advanced to the target site: (a) Through the vascular system, where a catheter is advanced from the femoral vein/artery or any other blood vessel that allows access to the target site; (b) Trans-apically, where a catheter is advanced through a small incision made in the chest wall and then through the apex; or (c) Trans-atrially, where a catheter is advanced through a small incision made in the chest wall and then through the left or right atrium.

SUMMARY

Some embodiments disclosed herein are directed towards an annuloplasty ring for heart valve repair. In some embodiments, the annuloplasty ring may have a first end and a second end opposite the first end. The annuloplasty ring may be configured to have an elongate insertion geometry and an annular operable geometry. The annuloplasty ring may further comprise a plurality of anchors within the annuloplasty ring. The plurality of anchors may have a deployment configuration in the annular operable geometry. The annu-

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loplasty ring may comprise at least one synthetic leaflet in mechanical communication with the annuloplasty ring at a plurality of points. In some embodiments, the at least one synthetic leaflet may comprise a valve frame having a proximal opening and a synthetic leaflet material mechanically coupled to the valve frame. The synthetic leaflet material may be located within the proximal opening. In some embodiments, the valve frame may comprise a shape memory metal. In further embodiments, the synthetic leaflet material may comprise a polymeric material.

In some embodiments, the at least one synthetic leaflet of the annuloplasty ring may comprise a plurality of leaflets. In further embodiments, the plurality of leaflets may have a first leaflet positioned adjacent to the first end of the annuloplasty ring and a second leaflet positioned adjacent to the second end of the annuloplasty ring. In some embodiments, at least a portion of the first leaflet and the second leaflet are mechanically coupled.

In some embodiments, the at least one synthetic leaflet is in mechanical communication with the annuloplasty ring at a plurality of points on a posterior side or an anterior side of a native valve.

In additional embodiments, the annuloplasty device may further comprise a plurality of anchor windows, a coating, a DACRON (i.e., polyethylene terephthalate) coating, a shape memory metal, or a combination thereof. In some embodiments, the at least one synthetic leaflet may comprise one or more lace holes. In further embodiments, the annuloplasty ring may further comprise at least one snap mechanism configured to connect the first end and the second end.

Further embodiments disclosed herein are directed towards an anchor device. The anchor device may comprise a leaflet anchor having an anchor body with a first end and a second end. The leaflet anchor may further comprise an anchor portion in mechanical communication with the first end of the leaflet anchor. In some embodiments, the leaflet anchor may further comprise a separating element in mechanical communication with the second end of the leaflet anchor, and a distal portion in mechanical communication with the anchor body between the first end and the second end. In some embodiments, the separating element may be configured to rotate the anchor body. In further embodiments, the anchor portion or the anchor body may comprise a shape memory metal.

Additional embodiments disclosed herein are directed towards a method of delivering the annuloplasty ring in an elongate insertion geometry, where the delivery of the annuloplasty ring utilizes one of a trans-apical approach, trans-septal approach, a trans-femoral approach, a trans-jugular approach, and a trans-atrial approach.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects, features, benefits and advantages of the embodiments described herein will be apparent with regard to the following description, appended claims, and accompanying drawings where:

FIG. 1 depicts an annuloplasty ring having at least one synthetic leaflet in accordance with an embodiment.

FIG. 2 illustrates an annuloplasty ring in an open configuration having at least one synthetic leaflet in accordance with an embodiment.

FIG. 3 depicts a cross-sectional view of an annuloplasty ring in an elongate insertion geometry in accordance with an embodiment.

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FIG. 4 illustrates a perspective view of an annuloplasty ring having at least one synthetic leaflet in accordance with an embodiment.

FIG. 5A is a perspective view depicting delivery of an annuloplasty ring having at least one synthetic leaflet in accordance with an embodiment.

FIG. 5B illustrates a side view of an annuloplasty ring having at least one synthetic leaflet in accordance with an embodiment.

FIG. 5C is a perspective view depicting delivery of an annuloplasty ring having at least one synthetic leaflet in accordance with an embodiment.

FIG. 6 is a perspective view of an annuloplasty ring having at least one synthetic leaflet in an open configuration in accordance with an embodiment.

FIG. 7 depicts an annuloplasty ring having at least one anterior synthetic leaflet and at least one posterior synthetic leaflet in accordance with an embodiment.

FIG. 8 is a cross-section view of an annuloplasty ring having at least one anterior synthetic leaflet and at least one posterior synthetic leaflet in accordance with an embodiment.

FIG. 9 is a top view of an annuloplasty ring having at least one anterior synthetic leaflet and at least one posterior synthetic leaflet secured to a native leaflet with a leaflet anchor in accordance with an embodiment.

FIG. 10 is a side view of an annuloplasty ring having at least one synthetic leaflet in accordance with an embodiment.

FIG. 11A illustrates a leaflet anchor in accordance with an embodiment.

FIG. 11B depicts a deployed leaflet anchor in accordance with an embodiment.

FIG. 11C illustrates a deployed leaflet anchor in accordance with a further embodiment.

FIG. 12 is a top view of an annuloplasty ring having at least one posterior synthetic leaflet in accordance with an embodiment.

FIG. 13 depicts an annuloplasty ring having a plurality of synthetic leaflets in accordance with an embodiment.

DETAILED DESCRIPTION

The term “about,” as used herein, refers to variations in a numerical quantity that can occur, for example, through measuring or handling procedures in the real world; through inadvertent error in these procedures; through differences in the manufacture, source, or purity of compositions or reagents; and the like. Typically, the term “about” as used herein means greater or lesser than the value or range of values stated by $\frac{1}{10}$ of the stated values, e.g., $\pm 10\%$. The term “about” also refers to variations that would be recognized by one skilled in the art as being equivalent so long as such variations do not encompass known values practiced by the prior art. Each value or range of values preceded by the term “about” is also intended to encompass the embodiment of the stated absolute value or range of values. Whether or not modified by the term “about,” quantitative values recited in the claims include equivalents to the recited values, e.g., variations in the numerical quantity of such values that can occur, but would be recognized to be equivalents by a person skilled in the art.

The term “patient” and “subject” are interchangeable and may be taken to mean any living organism. As such, the terms “patient” and “subject” may include, but is not limited to, any non-human mammal, primate or human. In some embodiments, the “patient” or “subject” is a mammal, such

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as mice, rats, other rodents, rabbits, dogs, cats, swine, cattle, sheep, horses, primates, or humans. In some embodiments, the patient or subject is an adult, child or infant. In some embodiments, the patient or subject is a human.

When implanting a replacement valve (e.g., an aortic valve, mitral valve, tricuspid valve, or other valve), the replacement valve can include a prosthesis attachment. The prosthesis can be configured to secure the replacement valve in a patient’s heart. Additional detail related to prosthetic valves for mitral or tricuspid valve replacement can be found in: U.S. patent application Ser. No. 13/198,582 entitled “Percutaneous Transcatheter Repair of Heart Valves,” filed Aug. 4, 2011; U.S. patent application Ser. No. 13/397,545 entitled “Percutaneous Transcatheter Repair of Heart Valves via Trans-Apical Access,” filed Feb. 15, 2012; and U.S. patent application Ser. No. 13/779,478 entitled “Percutaneous Annuloplasty System with Anterior-Posterior Adjustment,” filed Feb. 27, 2013, the contents of which are incorporated herein by reference in their entirety.

An implant and delivery system for introduction of a semi-rigid ring for treatment of tricuspid or mitral valve regurgitation includes an annuloplasty ring comprising an outer hollow member with a plurality of segments. In a further embodiment, segments may be adjustable and may cooperate with one another in order to change the outer hollow member from an elongated insertion shaped geometry to an annular operable shaped geometry. The tricuspid annuloplasty ring may include one or more zones comprising internal anchor members located at least partially within the outer hollow member. In one non-limiting embodiment, the tricuspid annuloplasty ring may include up to four different anchor zones, which are further discussed herein. In an embodiment, the internal anchor members may be configured to emerge sequentially from windows (i.e., openings) along the hollow tube, thereby engaging the tissue of the tricuspid valve annulus under treatment, potentially in a predetermined sequence.

Systems and methods are provided for introducing an annuloplasty ring (e.g., while it is housed in a linear shape within the delivery system) in a trans-apical or trans-femoral approach. In an embodiment, the distal tip of the delivery system may be introduced above the annulus. Once the annuloplasty ring is introduced, the plane of the annuloplasty ring may be rotated (e.g., automatically) to be parallel to the plane of the annulus. Once in the proper location, an embodiment may deploy a plurality of anchors. For example, an embodiment may deploy anchors associated with the septal zone, the posterior zone, or the first or second anterior zones.

The annuloplasty ring may then be snapped into a proper shape (e.g., a “D” shape) and introduced to the stabilization tool. The shape is possible because, as discussed herein, the annuloplasty ring comprises an outer hollow member with a plurality of segments, where the segments may be adjustable and may cooperate with one another in order to change the outer hollow member from an annular operable shaped geometry to an elongated insertion shaped geometry and vice versa.

Referring to FIG. 1, a top view of an annuloplasty ring 100, as it relates to various embodiments discussed herein, is shown. As shown, the annuloplasty ring 100 may have an annular operable shaped geometry where a first end and a second end are in mechanical communication through snap mechanism 130. In an embodiment, the annuloplasty ring may have an operable geometry. For example, the hypotube may be annular and/or D-shaped. Additionally, FIG. 1 illustrates a plurality of anchors 120 deployed through a

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plurality of anchor windows **140** of the annuloplasty ring **100**. As shown in FIG. 1, the annuloplasty ring **100** may have at least one synthetic leaflet **200** in mechanical communication with the annuloplasty ring at a plurality of points on the inner surface of the annuloplasty ring **100**. In some embodiments, the synthetic leaflet **200** may comprise a valve frame **400** having a proximal opening. The at least one synthetic leaflet may further comprise a synthetic leaflet material mechanically coupled to the valve frame, where the synthetic leaflet material is within the proximal opening. As depicted in FIG. 1, the at least one synthetic leaflet may comprise a plurality of leaflets, where the plurality of leaflets are mechanically coupled.

The annuloplasty ring may be made of various materials (e.g., a shape memory hypotube (nickel titanium (Ni—Ti) super elastic alloy)) cut to form a plurality of segments. Additionally, the cutting pattern used for laser processing (e.g. the cutting of anchor windows **140** through which anchors **120** may be deployed) of the annuloplasty ring **100** is illustrated in FIG. 1.

In an embodiment, the valve frame **400** may be made of various materials including, but not limited to, a shape memory metal (e.g. Ni—Ti). The valve frame **400** may have a diameter in a range of about 0.01 inches to about 0.015 inches. The synthetic leaflet material **200** may be mechanically coupled to the valve frame **400** by suturing (e.g. surgical sutures **401**). This configuration may allow the at least one synthetic leaflet to move towards the surface of the annulus upon deployment of the annuloplasty ring. The suture material may be polytetrafluoroethylene (PTFE) or polypropylene sutures. Further embodiments may include a synthetic leaflet material made from a polymeric material (e.g. polyurethane or dried precordium tissue).

Referring to FIG. 2, a top view of the annuloplasty ring **100**, as it relates to various embodiments discussed herein, is shown. As shown, the annuloplasty ring **100** may have an elongate insertion geometry where the first end is opposite the second end. In some embodiments, the first end, the second end, or both may have a snap mechanism **130**. In some embodiments, the annuloplasty ring **100** may have a coating **500** (e.g. a DACRON coating) on the inner surface or the outer surface of the annuloplasty ring **100**. In further embodiments, the annuloplasty ring **100** may have at least one synthetic leaflet **200**. As depicted in FIG. 2, the at least one synthetic leaflet may comprise a plurality of leaflets positioned at the first end or the second end of the annuloplasty ring.

Illustrated in FIG. 3 is a cross section view of an annuloplasty ring in an elongate insertion geometry within a delivery catheter **600**.

FIG. 4 illustrates a side view of a distal end of the delivery catheter **600** with the annuloplasty ring **100** and the at least one synthetic leaflet **200** partially deployed from the distal end of the delivery catheter. Further illustrated in FIG. 4 is a pull-wire **700** that can be in mechanical communication with the annuloplasty ring **100**.

FIGS. 5A-C depict perspective views of the annuloplasty ring **100** where the plane of the annuloplasty ring **100** (in its annular operable geometry) has been changed to be perpendicular to the longitudinal axis of the delivery catheter **600**. As illustrated in FIG. 5A, the at least one synthetic leaflet **200** comprises a plurality of leaflets in mechanical communication. In some embodiments, the plurality of leaflets in mechanical communication are configured to be operably connected to the inner surface of the posterior side **900** of the annuloplasty ring **100**. In further embodiments, the plurality of leaflets in mechanical communication are configured to

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be operably connected to the inner surface of the anterior side **1000** of the annuloplasty ring **100**. As illustrated in FIG. 5B, in some embodiments annuloplasty ring **100** may comprise ring element **1100**.

FIG. 5C depicts a further embodiment wherein annuloplasty ring **100** comprises a plurality of leaflets having one or more lace holes **210**. The plurality of leaflets may be configured to be in mechanical communication using a suture lace **220** that operably connects the plurality of leaflets through the one or more lace holes **210**. In further embodiments, the suture lace **220** can be configured to pass through the snap mechanism **130** and into the delivery catheter **600**.

FIG. 6 depicts the deployed annuloplasty ring **100** and the deployed plurality of anchors **120** with at least one synthetic leaflet **200** tilted toward the annulus and a native posterior leaflet.

Referring to FIGS. 7-8, the annuloplasty ring **100**, as it relates to various embodiments discussed herein, is shown.

As shown, the annuloplasty ring **100** in an annular operable geometry (FIG. 7) or an elongate insertion geometry (FIG. 8) may comprise a plurality of leaflets **200**. In some embodiments, the inner surface of the posterior side of the annuloplasty ring **100** may comprise a plurality of leaflets having lace holes that provides mechanical communication for the plurality of leaflets. The plurality of leaflets may be in mechanical communication with the inner surface of the posterior side of the annuloplasty ring **100** at a plurality of points.

In a further embodiment, as illustrated in FIG. 12, the annuloplasty ring may comprise at least one synthetic leaflet on the inner surface of the posterior side of the annuloplasty ring **100**. The at least one synthetic leaflet may be in mechanical communication with the inner surface of the posterior side of the annuloplasty ring **100** at a plurality of points. Further, the snap mechanism **130** can be located on the anterior side of the annuloplasty ring **100**.

In a further embodiment, as illustrated in FIG. 13, the inner surface of the posterior side of the annuloplasty ring **100** may comprise a plurality of leaflets having lace holes that provide mechanical communication for the plurality of leaflets. The plurality of leaflets may be in mechanical communication with the inner surface of the posterior side of the annuloplasty ring **100** at a plurality of points. Further, the inner surface of the anterior side of the annuloplasty ring **100** may comprise at least one synthetic leaflet **200** in mechanical communication with the inner surface of the annuloplasty ring **100**. In some embodiments, as depicted in FIG. 13, the inner surface of the annuloplasty ring **100** may comprise a plurality of commissure leaflets **200a,b**.

In some embodiments, as illustrated in FIG. 9, a leaflet anchor **800** may be in mechanical communication with the at least one synthetic leaflet **200** of the annuloplasty ring **100**. In some embodiments, the leaflet anchor **800** may mechanically couple one or more native leaflets to the at least one synthetic leaflet **200**. This embodiment may allow the one or more synthetic leaflets **200** to co-apt, as illustrated in FIG. 10.

As illustrated in FIG. 11A-C, the leaflet anchor **800** may have an anchor body **830** with a first end and a second end. The leaflet anchor **800** may further comprise an anchor portion **820** in mechanical communication with the first end of the leaflet anchor. In further embodiments, the leaflet anchor **800** may also include a separating element **850** in mechanical communication with the second end of leaflet anchor. Separating element **850** can be configured to separate the anchor delivery catheter **835** and the leaflet anchor

800. The leaflet anchor **800** may also comprise a disk portion **840**, where the disk portion is in mechanical communication with the anchor body **830** between the first end and the second end. As illustrated in FIG. 11B, leaflet anchor **800** can be configured to deploy from delivery catheter **600**, from anchor delivery catheter **835**, or a combination thereof. In some embodiments, leaflet anchor **800** may be configured to pass through native leaflet **1200** and the at least one synthetic leaflet **200**, wherein anchor portion **820** is deployed on a surface of the at least one synthetic leaflet **200**. In some embodiments, the disk portion **840** can be configured to fix the anchor portion **820** to the at least one synthetic leaflet **200** and the native leaflet **1200**.

The annuloplasty ring as described above can be designed and shaped for various functions such as mitral valve replacement. A similar annuloplasty ring can be designed and constructed for tricuspid valve replacement as well. However, a tricuspid ring can be designed with additional features, such as a release zone positioned on the ring assembly at a location that will be adjacent to a patient's atrioventricular node or valves. In certain implementations, the release zone does not have any anchors. Rather, the alternate shape and profile of the release zone provides for interference between the annuloplasty ring and the patient's atrioventricular node or valves, thereby securing the ring assembly in position.

Some embodiments are directed towards a method of delivering the annuloplasty ring as substantially described above. In some embodiments, the annuloplasty ring may be delivered in an elongate insertion geometry, where the delivery of the annuloplasty ring utilizes one of a trans-apical approach, trans-septal approach, trans-femoral approach, trans-jugular approach, or a trans-atrial approach.

This disclosure is not limited to the particular apparatus, systems, devices and methods described, as these may vary. The terminology used in the description is for the purpose of describing the particular versions or embodiments only, and is not intended to limit the scope.

As used in this document, the singular forms "a," "an," and "the" include plural references unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art. Nothing in this disclosure is to be construed as an admission that the embodiments described in this disclosure are not entitled to antedate such disclosure by virtue of prior invention. As used in this document, the term "comprising" means "including, but not limited to."

In the detailed description above, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be used, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that various features of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

The present disclosure is not to be limited in terms of the particular embodiments described in this application, which are intended as illustrations of various features. Many modifications and variations can be made without departing from its spirit and scope, as will be apparent to those skilled in the

art. Functionally equivalent methods and apparatuses within the scope of the disclosure, in addition to those enumerated herein, will be apparent to those skilled in the art from the foregoing descriptions. Such modifications and variations are intended to fall within the scope of the appended claims. The present disclosure is to be limited only by the terms of the appended claims, along with the full scope of equivalents to which such claims are entitled. It is to be understood that this disclosure is not limited to particular methods, reagents, compounds, compositions or biological systems, which can, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

With respect to the use of substantially any plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

It will be understood by those within the art that, in general, terms used herein, and especially in the appended claims (for example, bodies of the appended claims) are generally intended as "open" terms (for example, the term "including" should be interpreted as "including but not limited to," the term "having" should be interpreted as "having at least," the term "includes" should be interpreted as "includes but is not limited to," et cetera). While various compositions, methods, and devices are described in terms of "comprising" various components or steps (interpreted as meaning "including, but not limited to"), the compositions, methods, and devices can also "consist essentially of" or "consist of" the various components and steps, and such terminology should be interpreted as defining essentially closed-member groups. It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present.

For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases "at least one" and "one or more" to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles "a" or "an" limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases "one or more" or "at least one" and indefinite articles such as "a" or "an" (for example, "a" and/or "an" should be interpreted to mean "at least one" or "one or more"); the same holds true for the use of definite articles used to introduce claim recitations.

In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (for example, the bare recitation of "two recitations," without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to "at least one of A, B, and C, et cetera" is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (for example, "a system having at least one of A, B, and C" would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, et cetera). In those instances where a convention analogous to "at least

one of A, B, or C, et cetera” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (for example, “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, et cetera). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

In addition, where features of the disclosure are described in terms of Markush groups, those skilled in the art will recognize that the disclosure is also thereby described in terms of any individual member or subgroup of members of the Markush group.

As will be understood by one skilled in the art, for any and all purposes, such as in terms of providing a written description, all ranges disclosed herein also encompass any and all possible subranges and combinations of subranges thereof. Any listed range can be easily recognized as sufficiently describing and enabling the same range being broken down into at least equal halves, thirds, quarters, fifths, tenths, et cetera. As a non-limiting example, each range discussed herein can be readily broken down into a lower third, middle third and upper third, et cetera. As will also be understood by one skilled in the art all language such as “up to,” “at least,” and the like include the number recited and refer to ranges that can be subsequently broken down into subranges as discussed above. Finally, as will be understood by one skilled in the art, a range includes each individual member. Thus, for example, a group having 1-3 cells refers to groups having 1, 2, or 3 cells. Similarly, a group having 1-5 cells refers to groups having 1, 2, 3, 4, or 5 cells, and so forth.

Various of the above-disclosed and other features and functions, or alternatives thereof, may be combined into many other different devices or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations or improvements therein may be subsequently made by those skilled in the art, each of which is also intended to be encompassed by the disclosed embodiments.

What is claimed is:

1. An annuloplasty device comprising:

an annuloplasty ring comprising a first end and a second end opposite the first end, the annuloplasty ring configured to transition between an elongate insertion geometry and an annular operable geometry;

a plurality of anchors disposed within the annuloplasty ring, the plurality of anchors configured to assume a deployment configuration when the annuloplasty ring is in the annular operable geometry; and

a plurality of leaflets in mechanical communication with and mechanically coupled directly to the annuloplasty ring at a plurality of points, at least one leaflet of the plurality of leaflets comprising (i) a valve frame comprising a proximal opening, and (ii) a synthetic leaflet material mechanically coupled to the valve frame, the synthetic leaflet material disposed within the proximal opening,

wherein the plurality of leaflets comprises a first leaflet positioned adjacent to the first end of the annuloplasty ring and a second leaflet positioned adjacent to the second end of the annuloplasty ring.

2. The annuloplasty device of claim 1, wherein the valve frame comprises a shape memory metal.

3. The annuloplasty device of claim 1, wherein the synthetic leaflet material comprises a polymeric material.

4. An annuloplasty device comprising:

an annuloplasty ring comprising a first end and a second end opposite the first end, the annuloplasty ring configured to transition between an elongate insertion geometry and an annular operable geometry;

a plurality of anchors disposed within the annuloplasty ring, the plurality of anchors configured to assume a deployment configuration when the annuloplasty ring is in the annular operable geometry; and

a plurality of leaflets in mechanical communication with and mechanically coupled directly to the annuloplasty ring at a plurality of points,

wherein the plurality of leaflets comprises a first leaflet positioned adjacent to the first end of the annuloplasty ring and a second leaflet positioned adjacent to the second end of the annuloplasty ring, at least a portion of each of the first leaflet and the second leaflet mechanically coupled to one another.

5. The annuloplasty device of claim 4, wherein the plurality of points are on a posterior side of a native valve.

6. The annuloplasty device of claim 4, wherein at least one leaflet of the plurality of leaflets is in mechanical communication with the annuloplasty ring at a subset of points of the plurality of points on an anterior side of the native valve.

7. The annuloplasty device of claim 4, wherein the annuloplasty ring further comprises a plurality of anchor windows through which the plurality of anchors are deployable.

8. The annuloplasty device of claim 4, wherein the annuloplasty ring comprises a coating.

9. The annuloplasty device of claim 8, wherein the coating comprises a polyethylene terephthalate coating.

10. The annuloplasty device of claim 4, wherein the annuloplasty ring comprises a shape memory metal.

11. The annuloplasty device of claim 4, wherein the annuloplasty ring further comprises at least one snap mechanism configured to connect the first end and the second end.

12. The annuloplasty device of claim 4, wherein at least one leaflet of the plurality of leaflets comprises a synthetic leaflet material.

13. An annuloplasty device comprising:

an annuloplasty ring comprising a first end and a second end opposite the first end, the annuloplasty ring configured to transition between an elongate insertion geometry and an annular operable geometry;

a plurality of anchors disposed within the annuloplasty ring, the plurality of anchors configured to assume a deployment configuration when the annuloplasty ring is in the annular operable geometry; and

a plurality of leaflets in mechanical communication with and mechanically coupled directly to the annuloplasty ring at a plurality of points, at least one leaflet of the plurality of leaflets comprising one or more lace holes configured to receive a suture therethrough,

wherein the plurality of leaflets comprises a first leaflet positioned adjacent to the first end of the annuloplasty ring and a second leaflet positioned adjacent to the second end of the annuloplasty ring.

14. A method of implanting an annuloplasty device within a patient, the method comprising:
delivering, into the patient, an annuloplasty ring in an elongate insertion geometry, the annuloplasty ring

comprising a plurality of leaflets in mechanical communication with and mechanically coupled directly to the annuloplasty ring at a plurality of points; and positioning the annuloplasty ring parallel to a plane of a valve annulus of the patient,

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wherein the plurality of leaflets comprise a first leaflet positioned adjacent to the first end of the annuloplasty ring and a second leaflet positioned adjacent to the second end of the annuloplasty ring, at least a portion of each of the first leaflet and the second leaflet mechanically coupled to one another.

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15. The method of claim **14**, wherein the annuloplasty ring is delivered using one of a trans-apical approach, a trans-septal approach, a trans-femoral approach, a trans-jugular approach, or a trans-atrial approach.

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16. The method of claim **14**, wherein at least one leaflet of the plurality of leaflets comprises:

a valve frame comprising a proximal opening; and

a synthetic leaflet material mechanically coupled to the valve frame, wherein the synthetic leaflet material is disposed within the proximal opening.

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17. The method of claim **16**, wherein the valve frame comprises a shape memory metal.

18. The method of claim **16**, wherein the synthetic leaflet material comprises a polymeric material.

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19. The method of claim **14**, wherein at least one leaflet of the plurality of leaflets comprises a synthetic leaflet material.

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